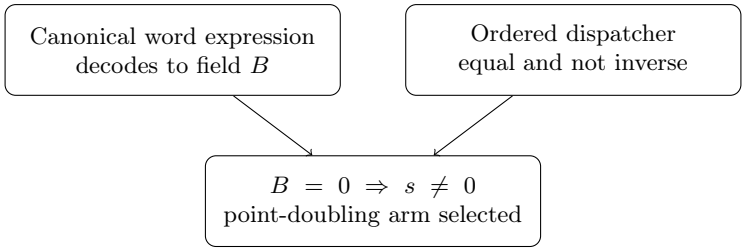
UInt64 ordered degree-two guard correspondence

Producer construction; independent source-fidelity review pending.

The word expression for B is canonical and decodes to

$$B = 2\big((qs - t)t - (rs)s\big).$$

The exact word dispatcher selecting `sameUDouble`, together with literal $B = 0$, implies $s \neq 0$. Thus its call to the helper cannot select `b_zero_identity`; it reaches the helper's point-doubling arm. This statement does not specify the degree or validity of the eventual doubled payload.



Domain and proof boundary

The modulus is prime, $5 \leq p < 2^{31}$; all twelve input coordinates are canonical. The modeled native route uses extension degree one and no lookup tables. The literal multiplication/subtraction association is preserved. The result reuses the reviewed field guard theorem and existing no-overflow word/classifier correspondence. It needs neither normalized Mumford validity nor nonzero weights: those are not asserted by this guard statement. The direct helper may still return identity outside this dispatch context.

Fixed controls

Coordinates below are (q, r, s, t, z, w), with each input compared to itself.

p and input	B	dispatch or boundary
$7 : (4, 0, 1, 4, 2, 3)$	0	sameUDouble; $s \neq 0$
$5 : (1, 0, 1, 1, 1, 1)$	0	sameUDouble; $s \neq 0$
$7 : (4, 0, 0, 0, 2, 3)$	0	sameUInverse; $s = 0$
$7 : (1, 0, 1, 1, 0, 1)$	0	sameUInverse despite equal inputs
$7 : (8, 0, 1, 1, 1, 1)$	—	noncanonical input
$p = 2$	—	excluded lower bound

Seven fixed Lean propositions include direct applications of the general theorem at $p = 5$ and $p = 7$. No model sweep, cohort run or cost measurement is included.

Evidence

Exact commands, source digests, before/after dependency bindings, failures and separate axiom readback accompany this report. The first proof attempt had redundant impossible-case handling; the first control attempt lacked an explicit decidability instance for the canonical-coordinate predicate. Both failures are retained. The final status is in `status.json`; typed fixtures are in `control-data.json`. Independent review precedes canonical integration.

Source: `projective_law_gate.rs`, lines 727–753 and 911–935; retained `PROOF_AUDIT.md`; reviewed `Guard22` and `CrossScaling` dependencies. Editable diagram: `flow.svg`.